

GPU and Model Compatibility Observatory

Project Proposal, Data Management 2025/2026

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Project Type	Data Warehousing, ROLAP (PostgreSQL)
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1. Datasets

The project integrates two well-scoped, statically downloadable datasets:

- Epoch AI, Notable AI Models Dataset (epoch.ai/data, CC-BY): a curated CSV of approximately 3,500 models spanning the history of AI from 1950 to 2026, with attributes including parameter count, training compute (FLOPs), training cost (USD), training hardware, organization, domain, and release date. This constitutes the primary model dimension of the data warehouse.
- Graphics Card Full Specs Dataset (Kaggle, kaggle.com/datasets/alanjo/graphics-card-full-specs): hardware specifications for consumer and datacenter GPUs, including VRAM (GB), memory bus width (bits), memory clock (MHz), and memory type (GDDR5/6/6X, HBM). Memory bandwidth in GB/s is not directly provided but is derived during ETL using the standard formula: $\text{bandwidth} = (\text{memClock} \times \text{memBusWidth} \times \text{dataRate}) / 8000$, where dataRate varies by memory type.

2. Description of the Work

The project addresses the following business question: given a specific GPU, which AI models can it realistically run, at what inference speed, and under what conditions? This is a concrete deployment feasibility problem faced by companies evaluating AI adoption without relying on cloud infrastructure.

The data warehouse is implemented in PostgreSQL following a ROLAP star schema. The central fact table, `FACT_GPU_MODEL_COMPATIBILITY`, stores one row per (GPU, model) pair and captures the following derived measures: `fits_in_vram` (boolean), `vram_headroom_gb` (numeric), `estimated_throughput` (numeric, NULL for non-generative models), `throughput_unit` (varchar, e.g. tokens/sec, images/sec, or NULL), and `quantization_required` (categorical: none, 4-bit, or 8-bit). The throughput estimate is derived from GPU memory bandwidth and model parameter count using the formula: $\text{throughput} = \text{memory_bandwidth} / (\text{parameters} \times \text{bytes_per_weight})$. This measure is meaningful only for generative models and is left NULL otherwise, reflecting the heterogeneous nature of the Epoch AI dataset, which spans LLMs, vision models, and historical architectures across seven decades. Dimension tables are: `DIM_MODEL` (parameter count, domain, organization, release year), `DIM_GPU` (manufacturer, VRAM, derived bandwidth, architecture, release year), and `DIM_TIME`.

ETL is performed by a Python script that loads both source CSVs, computes all derived attributes, and populates the star schema. The model dataset is used in its entirety (approximately 3,500 rows); the GPU dataset is filtered to cards released from 2016 onward, covering the modern deep learning hardware era and yielding approximately 500 GPU records. The resulting compatibility fact table contains on the order of 1.75 million rows, a tractable and well-defined scope.

OLAP analyses include: (i) how GPU deployability of state-of-the-art models has evolved over time as both model size and hardware capacity scaled; (ii) which GPU generations offer the best VRAM-to-deployable-model trade-off for modern LLMs; (iii) which AI domains (vision, language, multimodal) are most accessible without requiring quantization. Results are presented through a Next.js web application serving as a decision-support interface for non-technical company stakeholders, consistent with the data warehousing objective of making analytical findings accessible to business users.

The entire system runs locally via Docker Compose: a PostgreSQL container hosts the data warehouse, a Python ETL container performs the one-time data loading, and a Next.js container exposes the analytical interface. No cloud dependencies are required.

Dataset references: epoch.ai/data · kaggle.com/datasets/alanjo/graphics-card-full-specs